

JOYCE WONG TZE ENG

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EDUCATION

UNIVERSITI MALAYA

Bachelor of Computer Science & Information
Technology (Information System)

7/2026 • Current

- **Relevant Coursework:** Java, Mobile App Development, Web Development, Object-Oriented Programming, Human-Computer interaction, Data Analytics, Data Structures & Algorithms, Machine Learning

TECHNICAL SKILLS

- **Programming:** Python, Java, JavaScript, Dart, HTML/CSS
- **Frameworks:** Laravel, Flutter, React, Tailwind CSS, REST API
- **Data & Analytics:** pandas, scikit-learn, SQL, Power BI, Excel
- **Platforms & Tools:** Firebase, Git, Docker, Android Studio, Figma, Render, Netlify

PROJECTS

STITCHFLOW

April 2026

- Built an AI-powered document workflow engine as part of a team for UM Hackathon 2026, focused on human-in-the-loop automation for compliance-sensitive decisions.
- Designed a LangGraph-based agent pipeline that redacts PII, screens recommendations against policy rules, and pauses for human approval before execution, backed by a FastAPI backend and a tamper-evident, hash-chained audit log.

EXPENSE SPLITTER

May 2026

- Independently developed a Flutter-based expense-splitting app and took it through to a public web deployment.
- Resolved real deployment issues — renderer configuration, case-sensitive asset paths, and web-specific image picker behavior — and shipped the final build to Firebase Hosting.

CRAVE DROP — SMART FOOD DELIVERY SYSTEM

June 2026

- Collaborated in a 5-person team to build a food delivery and order management system for a Data Structures & Algorithms course, owning the Logistics & Rider Assignment module.
- Implemented a Min-Heap priority-queue engine for rider dispatch, reducing assignment lookup from linear-scan $O(n)$ to $O(\log n)$, and involved in viva-preparation Q&A materials for the full team.
- Independently built and deployed a web-based demo client for the platform, hosted live at cravedrop-web-demo.netlify.app.

ENERGY MARKET ANALYTICS DASHBOARD

July 2026

- Built an independent trading-analytics project, covering Brent Crude, WTI Crude, Natural Gas and USD/MYR.
- Developed a full data pipeline in Python (pandas, yfinance) computing a rebased-to-100 performance index, 20-day rolling volatility, and cross-market correlation, deployed as both an interactive web dashboard and a parallel Power BI/DAX model.

ACTIVITIES

UM HACKATHON 2026

Participant

April 2026

- Built StitchFlow, an AI-powered document workflow engine, as part of a 4-person team within a week hackathon timeframe.
- Involved in both frontend and backend development, implementing a React/TypeScript interface alongside a FastAPI backend with a LangGraph-based human-in-the-loop agent pipeline.
- Collaborated closely with teammates to integrate frontend and backend components under time pressure, ensuring a working end-to-end demo by submission deadline.

Languages

English, Chinese, Malay